Certified Feature Selection for Wasserstein Robust Hypothesis Testing

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Abstract

We study budgeted feature selection for binary hypothesis testing under class-conditional Wasserstein uncertainty. For finite, nonnegative, nondecreasing convex losses, an exact finite formulation constructs a whole-space test from every feasible solution. Scaling coordinate transport prices by the risk gives dual certificate fractions that are exact on binary selections. Every chosen subset receives a full continuous solve, producing an actual feasible-test upper bound and a globally valid tangent. A cheap Gaussian-process predictor uses completed solves to rank a small binary pool; a global integer master supplies the certified lower bound. Predictions determine which subset to try, but never replace an inner solve or provide certificates. Verified solve accuracies imply finite termination at the requested risk precision, with an evaluation bound determined by the data movement weights. Population coverage requires separate calibration.

1 The selection problem

We can observe at most d of D features. The choice of features and the test using them must be optimized together: a good subset is one on which a test can keep both classwise losses small when the distributions move within their specified uncertainty sets.

Pool the observations from the two classes into $x_1, \dots, x_N \in \mathbb{R}^D$. Observation i has class $c_i \in \{0,1\}$ and weight $q_i > 0$. We encode its class by the sign $y_i = 2c_i - 1$: $y_i = -1$ for class zero and $y_i = +1$ for class one. Both c_i and y_i are fixed, observed labels, not optimization variables. Write $I_k = \{i : c_i = k\}$. The empirical distributions and uncertainty sets are

$$\widehat{P}_k = \sum_{i \in I_k} q_i \delta_{x_i}, \qquad \sum_{i \in I_k} q_i = 1, \qquad \mathcal{U}_k = \{P : W_1(P, \widehat{P}_k) \le \theta_k\}, \quad k = 0, 1.$$

Here δ_x denotes unit mass at x. The distance W_1 is the minimum expected L^1 distance needed to transport one distribution to the other; the distributions have finite first moments. The radii $\theta_k > 0$ and feature scales are fixed before selection. Each class has total weight one, so the pooled weights sum to two.

For an integer budget $0 \le d \le D$, a binary vector z selects $S(z) = \{j : z_j = 1\}$, with $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}$. A measurable score g depends only on the selected coordinates. A positive score favors class zero; the losses are therefore $\Phi(-g)$ for class zero and $\Phi(g)$ for class one. The problem is

$$\begin{aligned} V(z) &= \inf_{g = g(x_{S(z)})} \left\{ \sup_{P \in \mathcal{U}_0} \mathbb{E}_P \Phi(-g) + \sup_{P \in \mathcal{U}_1} \mathbb{E}_P \Phi(g) \right\}, \\ \mathrm{OPT} &= \min_{z \in \mathcal{Z}_d} V(z). \end{aligned} \tag{1}$$

Thus $V(z)$ already includes the best possible test for the chosen features; selection does not hold a previously fitted test fixed.

The function $\Phi : \mathbb{R} \to [0, \infty)$ is finite, convex and nondecreasing. Nondecreasing means that a larger score against the true class cannot reduce its loss. We consider nonconstant losses with $\Phi(0) > 0$. If $\Phi(0) = 0$ or the loss is constant, the constant test already solves the problem. Specific formulas for Φ are not needed for the construction below; examples are given in Section 6. The constant score $g = 0$ provides the common benchmark

$$0 \le V(z) \le B := 2\Phi(0).$$

2 A finite problem that still designs the whole test

The same pooled observations serve as candidate target locations. To compare source i with target ℓ, use one vector of coordinatewise distances:

$$\mathbf{d}_{i\ell} = |x_i - x_\ell| \in \mathbb{R}^D_+, \qquad d_z(x_i, x_\ell) = \mathbf{d}_{i\ell}^\top z.$$

Absolute value is taken coordinatewise. The vector stores the geometry once; its inner product with z is the distance using the selected features.

Let u_ℓ be the score assigned to target ℓ. Let a_i be a loss allowance at source i, and let $\lambda_k \geq 0$ be the price per unit of transport for class k. Write $q = (q_i)$, $\theta = (\theta_0, \theta_1)^\top$, $\lambda = (\lambda_0, \lambda_1)^\top$, and $J(a, \lambda) = q^\top a + \theta^\top \lambda$.

Theorem 1 (Exact finite formulation)**.** *For every binary z,*

$$\begin{aligned} V(z) = \min_{a,u,\lambda} \quad & J(a,\lambda) \\ \textit{subject to} \quad & \Phi(y_i u_\ell) \leq a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z, \qquad i, \ell = 1, \dots, N, \\ & \lambda \geq 0, \qquad a, u \in \mathbb{R}^N. \end{aligned} \tag{2}$$

The minimum is attained. Every feasible solution yields a measurable test on the entire observation space, using only the selected coordinates, with robust risk at most $J(a, \lambda)$.

The inequality in (2) has a direct meaning: after moving from source i to target ℓ, the loss is at most its source allowance plus the transport price. The source retains its class c_i and sign y_i, while u_ℓ is the score at the target. Explicitly,

$$\Phi(y_i u_\ell) = \begin{cases} \Phi(-u_\ell), & c_i = 0, \\ \Phi(u_\ell), & c_i = 1. \end{cases}$$

Thus the loss uses the source class even when the target observation originally belongs to the other class; either class can send mass to any target location.

The whole-space conclusion is essential. It allows a finite feasible objective to be used as an upper bound for the original test problem. Its proof blends two source scores, one from each class. Convexity controls the blended losses, and the triangle inequality controls the required transport allowances. The construction and both directions of the equality are given in Appendix A; no particular inverse or closed form of Φ is used.

3 Use the risk to control which coordinates are available

Taking $\ell = i$ in (2) gives $a_i \geq \Phi(y_i u_i) \geq 0$. Consequently,

$$\theta^\top \lambda \leq J(a, \lambda). \tag{3}$$

The transport-price contribution is part of the total risk, so it cannot exceed that risk. This inequality supplies the bound needed to connect feature selection with continuous test design.

Introduce two vectors $w_0, w_1 \in \mathbb{R}^D_+$ of coordinate prices. On a binary selection we can take $w_{kj} = \lambda_k z_j$: an unobserved coordinate has price zero. To obtain a useful continuous extension, use the risk bound η to constrain these prices:

$$\begin{aligned} v(z) = \min_{\eta,a,u,\lambda,w_0,w_1} \quad & \eta \\ \text{subject to} \quad & \Phi(y_i u_\ell) \leq a_i + \mathbf{d}_{i\ell}^\top w_{c_i}, \qquad \forall i, \ell, \\ & J(a,\lambda) \leq \eta, \qquad \lambda \geq 0, \\ & 0 \leq w_k \leq \lambda_k \mathbf{1}, \qquad k = 0, 1, \\ & \theta_0 w_0 + \theta_1 w_1 \leq \eta z. \end{aligned} \tag{4}$$

Vector inequalities hold coordinatewise; a, u, η are real variables. The first line designs the test using the coordinate prices. The next lines bound its risk and keep each coordinate price below the class price. The last line makes an unselected coordinate unavailable and ties the allowed prices to the risk being minimized.

For binary z, an optimizer of (2) enters (4) with $\eta = J$ and $w_k = \lambda_k z$, by (3). Conversely, a feasible point of (4) has zero prices on unselected coordinates and $w_k \le \lambda_k z$. The same a, u, λ therefore satisfy (2), with objective at most η. Hence $v(z) = V(z)$ at every binary selection. For fractional $z \in [0,1]^D$, (4) defines the extension used to construct lower bounds.

Fixing z makes (4) a convex problem. This alone does not prove that its value $v(z)$ is convex in z, since the last line contains the product ηz. The next step establishes that property and supplies information that remains useful when the feature subset changes.

4 One solved subset gives lower bounds for all subsets

Consider a nonnegative matrix Π with $\Pi_{i\ell}$ units of mass sent from source i to target ℓ. Its rows retain the source weights, $\Pi\mathbf{1} = q$. The masses received from the two classes and their coordinatewise movements are

$$p_{k\ell} = \sum_{i \in I_k} \Pi_{i\ell}, \qquad A_k(\Pi) = \sum_{i \in I_k, \ell} \Pi_{i\ell} \mathbf{d}_{i\ell} \in \mathbb{R}^D_+.$$

At a target receiving both classes, one score must serve both. The least weighted loss that this common score can achieve is

$$\Psi(p_0, p_1) = \inf_s \{p_0 \Phi(-s) + p_1 \Phi(s)\}, \qquad H(\Pi) = \sum_\ell \Psi(p_{0\ell}, p_{1\ell}). \tag{5}$$

The infimum permits limiting scores when one class has zero mass. H measures the loss left after the best score is chosen at every target. It is nonnegative. Scaling all masses by $t \ge 0$ scales H by t; this property concerns the masses and does not require Φ itself to be homogeneous.

To use a matrix that may move more mass than the original budget allows, introduce a nonnegative correction vector $\mu \in \mathbb{R}^D_+$. The pair (Π, μ) must satisfy

$$\Pi \ge 0, \qquad \Pi\mathbf{1} = q, \qquad \mu \ge 0, \qquad \mathbf{1}^\top \big(A_k(\Pi) - \theta_k \mu\big)_+ \le \theta_k, \quad k = 0, 1. \tag{6}$$

Here $t_+ = \max(t, 0)$, applied coordinatewise to a vector. After $\theta_k \mu_j$ is deducted from movement in coordinate j, the remaining positive movements must fit inside the class budget θ_k. These conditions contain no z: the same pair can be used at any selection. The matrix need not be a transport plan inside the original uncertainty balls. The correction is what makes it useful as a risk lower bound.

The reason for the correction in the denominator can be seen directly. For $0 \le z \le 1$, (6) implies $z^\top A_k(\Pi) \le \theta_k(1 + \mu^\top z)$. For any feasible point of (2) on a binary selection,

$$\begin{aligned} H(\Pi) &\le \sum_{i,\ell} \Pi_{i\ell} \Phi(y_i u_\ell) \\ &\le q^\top a + \sum_k \lambda_k z^\top A_k(\Pi) \\ &\le J(a, \lambda) + (\mu^\top z) \theta^\top \lambda \\ &\le (1 + \mu^\top z) J(a, \lambda). \end{aligned}$$

The four steps use, respectively, best-score loss, finite loss constraints, the corrected transport budgets, and (3). Thus $H(\Pi)/(1 + \mu^\top z)$ is a lower bound on $V(z)$.

Theorem 2 (Exact representation and convexity)**.** *For every* $z \in [0,1]^D$,

$$v(z) = \max_{\Pi,\mu \text{ satisfying (6)}} \frac{H(\Pi)}{1+\mu^\top z}. \tag{7}$$

The maximum is attained, v *is convex, and* $v(z) = V(z)$ *on binary selections.*

The preceding inequalities explain why each fraction is a valid lower bound. Equality with v requires more: minimizing the Lagrangian of (4) gives the row masses, transport inequalities and one normalization equation. Rescaling those multipliers gives exactly (7). Appendix B derives the objective and every constraint by eliminating one group of variables at a time. For fixed (Π, μ), the fraction is convex because its numerator is nonnegative and its denominator is a positive affine function. The maximum of these convex lower functions is therefore convex.

5 Choosing which binary subset to evaluate

Fix a requested risk accuracy $\varepsilon > 0$. Let $\delta \le \varepsilon/4$ be the allowed inner risk gap, and $\xi \le \varepsilon/4$ the allowed master candidate gap, both nonnegative. Previous evaluations help choose the next binary subset. Every chosen subset then receives a full solve of (4), with all continuous variables reoptimized.

Solve the subset and store a global cut. At $z^s \in \mathcal{Z}_d$, the inner solve returns a finite feasible solution (a^s, u^s, λ^s) and its whole-space test. Its verified risk bound is $R_s = J(a^s, \lambda^s)$. Retaining the constant test ensures $R_s \le B$. The dual solve returns a pair (Π^s, μ^s) satisfying (6). Write its verified fraction as $f_s(z) = h_s/[1 + (\mu^s)^\top z]$, with $0 \le h_s \le H(\Pi^s)$. The certified value $\ell_s \le f_s(z^s)$ must satisfy

$$\ell_s \le V(z^s) \le R_s, \qquad 0 \le R_s - \ell_s \le \delta \le \varepsilon/4.$$

In exact arithmetic $\ell_s = f_s(z^s)$; any numerical downward shift is included in this gap. The fraction's gradient gives the cut

$$g_s = -\frac{h_s \mu^s}{[1 + (\mu^s)^\top z^s]^2}, \qquad L_s(z) = \ell_s + g_s^\top (z - z^s) \le v(z). \tag{8}$$

Convexity puts the tangent below f_s, while (7) puts f_s below v. Store this cut and the actual feasible test together.

Rank a small set of binary proposals. After t full evaluations, the actual test bounds and retained cuts give

$$U_t = \min\{B, R_1, \dots, R_t\}, \qquad a_t(z) = \max\{0, L_1(z), \dots, L_t(z)\}.$$

Here U_t is the global upper bound and $a_t(z)$ is a lower estimate of each subset's optimal risk. For $t = 0$, set $U_0 = B$ and $a_0 = 0$.

Fit a cheap Gaussian-process predictor to the observed pairs (z^s, R_s). A fixed similarity between subsets lets an evaluated subset inform nearby subsets, which receive lower predictive uncertainty when more strongly constrained by these observations. Denote the predicted risk by $m_t(z)$ and its predictive standard deviation by $\sigma_t(z)$. Both are used for ranking, not as guaranteed risk bounds; their construction is given in Appendix E.

Form a small finite pool $\mathcal{C}_t \subseteq \mathcal{Z}_d$ using a few coordinate swaps and random proposals. Keep subsets whose lower estimate allows sufficient improvement, and choose

$$z^{t+1} \in \operatorname*{argmin}_{\substack{z \in \mathcal{C}_t \\ a_t(z) \le U_t - \varepsilon/2}} \big[m_t(z) - \sigma_t(z)\big].$$

The score favors low predicted risk while allowing uncertain subsets to be tried. The screen exceeds the inner error $\delta \le \varepsilon/4$, so a resolved subset cannot be chosen again. The chosen subset is then fully solved: only its actual R_s can update U_t, and only its verified dual result supplies a new cut. The predictor does neither.

Use the global master to certify stopping. The master minimizes the lower model over all allowed subsets:

$$\min_{z\in Z_d,\eta} \eta \quad \text{subject to} \quad \eta \ge 0, \qquad \eta \ge L_s(z) \quad (s = 1, \dots, t). \tag{9}$$

It returns a binary z_{M} and a certified global lower bound. Update LB_t with the maximum of its previous value and this bound; all earlier cuts remain. Stop if $U_t - \mathrm{LB}_t \le \varepsilon$. Otherwise require $a_t(z_{\mathrm{M}}) - \mathrm{LB}_t \le \xi \le \varepsilon/4$ after the update. Then

$$a_t(z_{\mathrm{M}}) \le \mathrm{LB}_t + \xi < U_t - \varepsilon + \xi \le U_t - 3\varepsilon/4,$$

so the master supplies an eligible candidate. Refresh it every three full evaluations and whenever the pool has no eligible subset. If a solve cannot attain its required accuracy within the budget, return the available bounds and test with that status.

Algorithm 1 Full inner solves with prediction-based candidate selection

Require: $\varepsilon > 0$; $0 \le \delta \le \varepsilon/4$; $0 \le \xi \le \varepsilon/4$.

Set $t = 0$, $U_0 = B$, $\mathrm{LB}_0 = 0$; retain the constant test and initialize empty histories and the predictor.
while $U_t - \mathrm{LB}_t > \varepsilon$ **do**
  Form the small binary pool $\mathcal{C}_t$ and keep proposals with $a_t(z) \le U_t - \varepsilon/2$.
  if no proposal is eligible, or three full evaluations have occurred since the last master solve **then**
    Solve (9) and update LB_t with its certified bound.
    If $U_t - \mathrm{LB}_t \le \varepsilon$, return the retained best test; otherwise require $a_t(z_{\mathrm{M}}) - \mathrm{LB}_t \le \xi$ and add z_{M} to the eligible pool.
  end if
  Choose an eligible proposal minimizing $m_t(z) - \sigma_t(z)$.
  Set $s = t + 1$, $z^s = z$; fully solve (4) there, obtaining a feasible test and a verified fraction with $R_s - \ell_s \le \delta$.
  Compute (8) at this evaluated point; store its cut, subset and actual test bound; retain every previous cut.
  Update the predictor with the completed observation (z^s, R_s).
  Set $U_{t+1} = \min\{U_t, R_s\}$, $\mathrm{LB}_{t+1} = \mathrm{LB}_t$, and $t \leftarrow t + 1$.
end while
return The retained best test and subset, with interval $[\mathrm{LB}_t, U_t]$.

The evaluation bound uses the coordinate movement in the data. Let $d_{i\ell,j}$ be entry j of $\mathbf{d}_{i\ell}$ and define

$$M_j = \max_{k=0,1} \frac{\sum_{i\in I_k} q_i \max_\ell d_{i\ell,j}}{\theta_k}, \qquad \rho(z, z') = \sum_j M_j |z_j - z'_j|.$$

Thus ρ measures how selections differ in transport units. Clipping a feasible correction to $0 \le \mu^s \le M$ preserves feasibility; use it before computing the value and gradient in (8). The positive separation scale and number of retained coordinate weights are

$$r_\varepsilon = \frac{\varepsilon/2 - \delta}{B} > 0, \qquad M_{(1)} \ge \cdots \ge M_{(D)}, \qquad s_\varepsilon = \min\left\{ s \in \{0, \dots, D\} : \sum_{j>s} M_{(j)} < r_\varepsilon \right\}.$$

The proof shows that evaluated subsets are at least r_ε apart in ρ; s_ε makes the omitted weight smaller than this distance. Let T_ε count completed full binary evaluations begun before stopping, including the first. Refinements at one selected point count as one evaluation.

Theorem 3 (Global optimality and evaluation bound)**.** *Algorithm 1, with the stated feasibility, correction and accuracy requirements and all generated cuts retained, maintains*

$$\mathrm{LB}_t \le \mathrm{OPT} \le U_t. \tag{10}$$

If the required solves complete without a premature budget exit, it returns a feasible test with robust risk at most $\mathrm{OPT} + \varepsilon$*. The number of completed evaluations satisfies*

$$T_\varepsilon \le \min\left\{|\mathcal{Z}_d|, \sum_{i=0}^{\min(d,s_\varepsilon)} \binom{s_\varepsilon}{i}\right\} \le 2^{s_\varepsilon}.$$

Appendix C proves these statements. The predictor affects which eligible subset is tried, not the validity of the bounds or the count.

6 Loss choices and the scope of the result

The loss enters through Φ in the continuous problem and through Ψ in the dual objective. Examples covered by the same derivation are shown below; boundary values use continuous limits.

Loss	$\Phi(s)$	$\Psi(p_0, p_1)$
Half hinge	$\frac{1}{2}(1+s)_+$	$\min(p_0, p_1)$
Logistic	$\log(1+e^s)$	$(p_0+p_1)\log(p_0+p_1) - p_0\log p_0 - p_1\log p_1$
Exponential	e^s	$2\sqrt{p_0 p_1}$
Squared hinge	$(1+s)_+^2$	$4p_0p_1/(p_0+p_1)$

For example, half-hinge masses $p_0 = 0.2$ and $p_1 = 0.5$ leave loss 0.2: the best common score favors the larger mass but cannot distinguish the smaller mass at the same location. Appendix D derives all four formulas. For half hinge, a score clipped to $[-1, 1]$ gives randomized class-zero probability $(1 + g)/2$, so $V/2$ is the equal-prior worst-case error. Other losses define surrogate objectives. The current implementation uses half hinge and logistic; the other rows illustrate the theoretical scope.

The loss formulas need not be fixed before deriving the algorithm. The transport geometry still matters: the whole-space construction uses the triangle inequality, and the coordinate-price formulation uses the L^1 decomposition. The certified interval concerns the stated robust optimization problem; whether its uncertainty sets cover an unknown population is a separate statistical question.

A Proof of Theorem 1: the finite formulation

Under the assumptions of Section 1, let $T(z)$ be the infimum of the finite program in (2). Equality with $V(z)$ requires two directions. Restricting adversarial transport to the observed target pool gives $T(z) \le V(z)$. For the reverse direction, we extend any finite feasible solution to a whole-space test with the same loss allowances, giving $V(z) \le T(z)$. The second direction explains why solving the finite problem is sufficient to construct a test. We prove these inequalities first, then establish attainment.

A.1 Restrict the adversary, then identify the finite objective

Fix binary z and $S = \{j : z_j = 1\}$. Write P_S for the distribution of the selected coordinates under P. A transport in the selected coordinates can be realized in the original space by leaving each source's unselected coordinates fixed. The following identities verify its cost and test loss:

$$\begin{aligned} d_z(x, x') &:= \sum_{j=1}^{D} z_j |x_j - x'_j| = \|x_S - x'_S\|_1 \le \|x - x'\|_1, \\ W_1(P_S, \widehat{P}_{k,S}) &\le W_1(P, \widehat{P}_k), \\ \|x_i - (x_{\ell,S}, x_{i,S^c})\|_1 &= d_z(x_i, x_\ell), \\ g(x_{\ell,S}, x_{i,S^c}) &= g(x_\ell). \end{aligned} \tag{A.1}$$

Lifting retains the unselected source coordinates, so it preserves both selected transport cost and test loss. If source projections coincide, split their transported mass according to their weights and retain their source identities. The same construction lifts arbitrary projected targets and preserves finite first moments.

For scores $u = (u_\ell)_{\ell=1}^N$, define $Q_k(u)$ as the largest class-k loss obtained by sending mass to the projected pool. The variable $\Pi_{i\ell}$ sends mass from source $i \in I_k$ to target ℓ; only class-k rows occur in this calculation:

$$\begin{aligned} Q_k(u) = \max_{\Pi \ge 0} \quad & \sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i u_\ell) \\ \text{subject to} \quad & \sum_{\ell} \Pi_{i\ell} = q_i \quad (i \in I_k), \\ & \sum_{i \in I_k, \ell} \Pi_{i\ell} d_z(x_i, x_\ell) \le \theta_k. \end{aligned} \tag{A.2}$$

For any original test, define $u(g) = (g(x_1), \dots, g(x_N))$. Lift each feasible transport using (A.1): its full-space distribution lies in $\mathcal{U}_k$ and its loss is $\sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i g(x_\ell))$. Thus

$$Q_k(u(g)) \le \sup_{P \in \mathcal{U}_k} \mathbb{E}_P \Phi((2k-1)g). \tag{A.3}$$

Staying at the source is feasible, and $0 \le \Pi_{i\ell} \le q_i$ makes the feasible set compact. To express this restricted worst-case loss using the finite primal variables, derive its LP dual. Assign unrestricted row multipliers a_i and a nonnegative budget multiplier λ_k. Because this is a maximization problem, the

Lagrangian is

$$\begin{aligned}\mathcal{L}_k &= \sum_{i\in I_k,\ell} \Pi_{i\ell}\Phi(y_i u_\ell) + \sum_{i\in I_k} a_i\left(q_i - \sum_{\ell}\Pi_{i\ell}\right)\\ &\quad + \lambda_k\left(\theta_k - \sum_{i\in I_k,\ell}\Pi_{i\ell} d_z(x_i,x_\ell)\right)\\ &= \sum_{i\in I_k} q_i a_i + \theta_k\lambda_k + \sum_{i\in I_k,\ell}\Pi_{i\ell}[\Phi(y_i u_\ell) - a_i - \lambda_k d_z(x_i,x_\ell)].\end{aligned} \tag{A.4}$$

For a feasible transport the row terms vanish and the budget term is nonnegative, so this Lagrangian bounds its loss above. After these constraints enter the Lagrangian, its supremum retains only $\Pi \geq 0$. A positive mass coefficient gives $+\infty$; if every coefficient is nonpositive, $\Pi = 0$ attains the supremum:

$$\sup_{\Pi\geq 0}\mathcal{L}_k = \begin{cases}\sum_{i\in I_k} q_i a_i + \theta_k\lambda_k, & \Phi(y_i u_\ell) \leq a_i + \lambda_k d_z(x_i,x_\ell) \quad (i\in I_k,\ \forall\ell),\\ +\infty, & \text{otherwise.}\end{cases} \tag{A.5}$$

Minimize this upper bound over $a_i \in \mathbb{R}$ and $\lambda_k \geq 0$. Finite linear programming strong duality gives

$$Q_k(u) = \min_{\substack{a_i\in\mathbb{R},\ \lambda_k\geq 0\\ \Phi(y_i u_\ell)\leq a_i+\lambda_k d_z(x_i,x_\ell)\ (i\in I_k,\ \forall\ell)}}\left[\sum_{i\in I_k} q_i a_i + \theta_k\lambda_k\right]. \tag{A.6}$$

Both classwise dual objectives use the same score vector u. Minimizing their sum over u is exactly the finite program (2), so $T(z) = \inf_u[Q_0(u) + Q_1(u)]$. Restricting the infimum to induced scores and using (A.3) gives

$$\begin{aligned}T(z) &= \inf_u[Q_0(u) + Q_1(u)]\\ &\leq \inf_g[Q_0(u(g)) + Q_1(u(g))] \leq V(z).\end{aligned} \tag{A.7}$$

Separate scores at coincident projected targets only relax the first infimum.

A.2 Extend the finite loss bounds to the whole space

Take a feasible (a,u,λ) in (2). We need a score g on the whole space whose losses satisfy

$$\Phi((2k-1)g(x)) \leq a_i + \lambda_k d_z(x_i,x) \quad (i\in I_k,\ k=0,1).$$

Integrating this inequality will turn the transport budget into the objective $J(a,\lambda)$. To satisfy every source inequality at once, it suffices to satisfy the smallest allowance in each class. Define

$$\begin{aligned}E_k(x) &= \min_{i\in I_k}\{a_i + \lambda_k d_z(x_i,x)\},\\ i_k(x) &= \min\arg\min_{i\in I_k}\{a_i + \lambda_k d_z(x_i,x)\}.\end{aligned} \tag{A.8}$$

Choose the smallest source index in a tie. At fixed x, abbreviate $i_k = i_k(x)$; c is the distance between the chosen sources and r_k their distances to x:

$$c = d_z(x_{i_0},x_{i_1}), \qquad r_k = d_z(x_{i_k},x), \qquad c \leq r_0 + r_1. \tag{A.9}$$

Substitute source–target pairs (i_0,i_0), (i_0,i_1), (i_1,i_0), and (i_1,i_1) in (2). Their source signs are -1 and $+1$, and the own-source distances are zero:

$$\begin{aligned}\Phi(-u_{i_0}) &\leq a_{i_0}, & \Phi(-u_{i_1}) &\leq a_{i_0} + \lambda_0 c,\\ \Phi(u_{i_0}) &\leq a_{i_1} + \lambda_1 c, & \Phi(u_{i_1}) &\leq a_{i_1}.\end{aligned} \tag{A.10}$$

Consider a convex combination with weight $\alpha \in [0,1]$. Convexity bounds the extra class-zero loss by $\lambda_0 \alpha c$ and the extra class-one loss by $\lambda_1(1-\alpha)c$. We therefore need $\alpha c \le r_0$ and $(1-\alpha)c \le r_1$. When $c > 0$, choose

$$\alpha = \frac{\min(c, r_0)}{c}, \qquad g(x) = (1-\alpha)u_{i_0} + \alpha u_{i_1}. \tag{A.11}$$

The triangle inequality ensures that both portions of this distance fit within r_0, r_1:

$$\begin{aligned} 0 \le \alpha \le 1, \qquad \alpha c &= \min(c, r_0) \le r_0, \\ (1-\alpha)c &= c - \min(c, r_0) = \max(c - r_0, 0) \le r_1. \end{aligned} \tag{A.12}$$

In particular, the last inequality uses $c - r_0 \le r_1$ and $r_1 \ge 0$.

Convexity, these four constraints, and (A.12) give the two pointwise loss bounds:

$$\begin{aligned} \Phi(-g(x)) &\le (1-\alpha)\Phi(-u_{i_0}) + \alpha\Phi(-u_{i_1}) \\ &\le (1-\alpha)a_{i_0} + \alpha(a_{i_0} + \lambda_0 c) = a_{i_0} + \lambda_0 \alpha c \\ &\le a_{i_0} + \lambda_0 r_0 = E_0(x), \\ \Phi(g(x)) &\le (1-\alpha)\Phi(u_{i_0}) + \alpha\Phi(u_{i_1}) \\ &\le (1-\alpha)(a_{i_1} + \lambda_1 c) + \alpha a_{i_1} = a_{i_1} + \lambda_1(1-\alpha)c \\ &\le a_{i_1} + \lambda_1 r_1 = E_1(x). \end{aligned} \tag{A.13}$$

If $c = 0$, set $g(x) = u_{i_0}$. The first and third endpoint inequalities give instead

$$\Phi(-g(x)) \le a_{i_0} \le E_0(x), \qquad \Phi(g(x)) \le a_{i_1} + \lambda_1 c = a_{i_1} \le E_1(x). \tag{A.14}$$

Finite comparisons of continuous functions make $i_k(x)$ Borel measurable, and hence the piecewise g is Borel measurable. It uses only x_S and lies between $\min_\ell u_\ell$ and $\max_\ell u_\ell$. It extends the loss bounds; interpolation of every prescribed u_ℓ is unnecessary.

The construction gives the required sourcewise loss bounds. Their expected source allowances are weighted by q_i, and their expected transport distance is at most θ_k. To make these two facts explicit, fix $P \in \mathcal{U}_k$ and choose a coupling of cost at most θ_k, retaining source indices and splitting coincident atoms according to their weights. Let γ_k be its measure on $I_k \times \mathbb{R}^D$, with target marginal P. Its source rows and cost satisfy

$$\gamma_k(\{i\} \times \mathbb{R}^D) = q_i, \qquad \int d_z(x_i, x)\, d\gamma_k(i,x) \le \int \|x_i - x\|_1\, d\gamma_k(i,x) \le \theta_k. \tag{A.15}$$

For each source $i \in I_k$, the constructed score satisfies $\Phi((2k-1)g(x)) \le E_k(x) \le a_i + \lambda_k d_z(x_i, x)$. Using both marginals gives

$$\begin{aligned} \mathbb{E}_P \Phi((2k-1)g) &= \int \Phi((2k-1)g(x))\, d\gamma_k(i,x) \\ &\le \int a_i\, d\gamma_k(i,x) + \lambda_k \int d_z(x_i, x)\, d\gamma_k(i,x) \\ &= \sum_{i \in I_k} a_i \gamma_k(\{i\} \times \mathbb{R}^D) + \lambda_k \int d_z(x_i, x)\, d\gamma_k(i,x) \\ &= \sum_{i \in I_k} q_i a_i + \lambda_k \int d_z(x_i, x)\, d\gamma_k(i,x) \\ &\le \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k. \end{aligned} \tag{A.16}$$

The bound is independent of P, so taking the classwise suprema and adding yields

$$\sum_{k=0}^{1} \sup_{P \in \mathcal{U}_k} \mathbb{E}_P \Phi((2k-1)g) \le \sum_{k=0}^{1} \left[\sum_{i \in I_k} q_i a_i + \theta_k \lambda_k \right] = J(a, \lambda).$$

Every finite feasible point therefore yields an original test with risk at most J. Taking the infimum gives $V(z) \le T(z)$; (A.7) gives the other direction. This proves the value equality and feasible-test claim in Theorem 1.

A.3 Attainment of the primal minima

To prove attainment for both (2) and (4), we bound all variables on a nonempty objective sublevel set and then use closedness.

The constant point $u = 0, a_i = \Phi(0), \lambda = 0$ is feasible in (2) with $J = B$. Adding $w_0 = w_1 = 0$ and $\eta = B$ makes it feasible in (4) for every $z \in [0,1]^D$. Thus the objective sublevels $J \le B$ and $J \le \eta \le B$ are nonempty. Own-source constraints give $a_i \ge \Phi(y_i u_i) \ge 0$; positive weights and radii then give

$$\begin{aligned} 0 \le q_i a_i \le q^\top a \le J \le B \quad &\Longrightarrow\ 0 \le a_i \le B/q_i, \\ 0 \le \theta_k \lambda_k \le \theta^\top \lambda \le J \le B \quad &\Longrightarrow\ 0 \le \lambda_k \le B/\theta_k, \\ 0 \le w_{kj} \le \lambda_k \le B/\theta_k, \quad 0 \le J \le \eta \le B \quad &\text{in (4).} \end{aligned} \tag{A.17}$$

Choose a fixed source $i_k \in I_k$ for each class. The finite pool distances and (A.17) bound both score signs by the finite constant

$$C = \max_{k,\ell} \left\{ B/q_{i_k} + (B/\theta_k) \|\mathbf{d}_{i_k \ell}\|_1 \right\},$$

$$\Phi((2k-1)u_\ell) \le B/q_{i_k} + (B/\theta_k) \|\mathbf{d}_{i_k \ell}\|_1 \le C.$$

We have bounded both $\Phi(u_\ell)$ and $\Phi(-u_\ell)$. To turn these loss bounds into a bound on the score itself, it remains to show that Φ grows without bound in the positive direction. Nonconstancy and monotonicity give $s_1 < s_2$ with $\Phi(s_2) > \Phi(s_1)$. Set $b = [\Phi(s_2) - \Phi(s_1)]/(s_2 - s_1) > 0$; convexity gives, for $t > s_2$,

$$\frac{\Phi(t) - \Phi(s_2)}{t - s_2} \ge b \quad \Longrightarrow \quad \Phi(t) \ge \Phi(s_2) + b(t - s_2) \longrightarrow \infty. \tag{A.18}$$

Choose $R > 0$ with $\Phi(R) > C$, possible by (A.18). Monotonicity gives

$$u_\ell \ge R \implies \Phi(u_\ell) > C, \qquad u_\ell \le -R \implies \Phi(-u_\ell) > C; \qquad |u_\ell| < R.$$

All variables on these sublevels are bounded. The finite convex loss is continuous, so the sublevels are closed and hence compact. Both continuous objectives attain their minima there. Since a constant test with value B is already feasible, restricting to these sublevels preserves both global minima.

B Proof of Theorem 2: from the risk model to global lower functions

Fix $z \in [0,1]^D$. We want to express the optimal risk $v(z)$ by pairs (Π, μ) that remain feasible when z changes. First minimize the Lagrangian over the test-design variables; the remaining multipliers describe masses, movements and a common scale τ. Dividing out this scale gives the fraction in (7). Binary equivalence then connects it to V, and convexity explains why its tangent is a global lower bound.

B.1 Eliminate the design variables to obtain the dual

In (4), the variables η, a, u are free and λ, w_0, w_1 are nonnegative. Attach one nonnegative multiplier to each inequality:

Constraint	Multiplier
$\Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i} \le 0$	$\bar{\Pi}_{i\ell}$
$J - \eta \le 0$	τ
$w_{kj} - \lambda_k \le 0$	$\bar{\beta}_{kj}$
$\theta_0 w_{0j} + \theta_1 w_{1j} - \eta z_j \le 0$	$\bar{\mu}_j$

Here $\bar{\Pi} \in \mathbb{R}_+^{N\times N}$, $\tau \ge 0$, and $\bar{\beta}_k, \bar{\mu} \in \mathbb{R}_+^D$. Bars indicate multipliers before their masses are normalized. Adding each multiplier times its constraint to the objective gives

$$\begin{aligned}\mathcal{L} = \eta + \tau(J-\eta) + \sum_{i,\ell} \bar{\Pi}_{i\ell}[\Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i}] \\ + \sum_k \bar{\beta}_k^\top (w_k - \lambda_k \mathbf{1}) + \bar{\mu}^\top(\theta_0 w_0 + \theta_1 w_1 - \eta z).\end{aligned} \tag{B.1}$$

At a primal feasible point, every added term is nonpositive. Thus fixed multipliers give the lower bound

$$\inf_{\substack{\eta\in\mathbb{R},\ a,u\in\mathbb{R}^N \\ \lambda, w_0, w_1 \ge 0}} \mathcal{L} \le \mathcal{L}(\eta, a, u, \lambda, w_0, w_1) \le \eta.$$

We maximize this lower bound after computing the infimum. The four constraint groups are already in $\mathcal{L}$ and are not imposed again; in particular, a and η remain free in this minimization.

To expose the coefficient of each design variable, collect the class masses and their movements:

$$\bar{p}_{k\ell} = \sum_{i\in I_k} \bar{\Pi}_{i\ell}, \qquad A_k(\bar{\Pi}) = \sum_{i\in I_k,\ell} \bar{\Pi}_{i\ell}\mathbf{d}_{i\ell}. \tag{B.2}$$

$$\begin{aligned}\sum_{i,\ell} \bar{\Pi}_{i\ell} a_i &= \sum_i a_i \sum_\ell \bar{\Pi}_{i\ell}, \\ \sum_{i,\ell} \bar{\Pi}_{i\ell}\Phi(y_i u_\ell) &= \sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)], \\ \sum_{i,\ell} \bar{\Pi}_{i\ell}\mathbf{d}_{i\ell}^\top w_{c_i} &= \sum_k w_k^\top A_k(\bar{\Pi}), \\ \tau J &= \sum_i \tau q_i a_i + \sum_k \tau\theta_k\lambda_k.\end{aligned} \tag{B.3}$$

Substitution separates $\mathcal{L}$ into five variable groups:

$$\begin{aligned}\mathcal{L} = \eta(1 - \tau - \bar{\mu}^\top z) + \sum_i a_i(\tau q_i - \sum_\ell \bar{\Pi}_{i\ell}) \\ + \sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)] \\ + \sum_k \lambda_k(\tau\theta_k - \mathbf{1}^\top\bar{\beta}_k) + \sum_k w_k^\top[\bar{\beta}_k + \theta_k\bar{\mu} - A_k(\bar{\Pi})].\end{aligned} \tag{B.4}$$

Now minimize each group. A linear term in a free variable has a finite infimum only when its coefficient is zero; a term in a nonnegative variable requires a nonnegative coefficient:

Here $b \in \mathbb{R}$ is a scalar coefficient and t a scalar variable.

$$\inf_{t\in\mathbb{R}} bt = \begin{cases} 0, & b = 0, \\ -\infty, & b \neq 0, \end{cases} \qquad \inf_{t\geq 0} bt = \begin{cases} 0, & b \geq 0, \\ -\infty, & b < 0. \end{cases}$$

Taking $n \to +\infty$, $t = -nb$ proves the first divergence when $b \neq 0$, and $t = n$ proves the second when $b < 0$; otherwise the infimum is zero at $t = 0$. Applying the first rule to η and a gives

$$\begin{aligned} \inf_{\eta\in\mathbb{R}} \eta(1 - \tau - \bar{\mu}^\top z) > -\infty \iff \tau + \bar{\mu}^\top z = 1, \\ \inf_{a\in\mathbb{R}^N} \sum_i a_i(\tau q_i - \sum_\ell \bar{\Pi}_{i\ell}) > -\infty \iff \bar{\Pi}\mathbf{1} = \tau q. \end{aligned} \tag{B.5}$$

The first equation normalizes the multipliers; the second fixes each source's row mass at τ times its original weight. The score variables contribute the best possible weighted loss at each target:

$$\begin{aligned} \inf_{u\in\mathbb{R}^N} \sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)] &= \sum_\ell \inf_{s\in\mathbb{R}} [\bar{p}_{0\ell}\Phi(-s) + \bar{p}_{1\ell}\Phi(s)] \\ &= \sum_\ell \Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) = H(\bar{\Pi}). \end{aligned} \tag{B.6}$$

This equality concerns infima and does not require a finite minimizing score. For any $\epsilon > 0$, choose each u_ℓ within ϵ/N of its scalar infimum:

$$\Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) \leq \bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell) \leq \Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) + \epsilon/N.$$

Summing and letting $\epsilon \downarrow 0$ proves (B.6), including zero class masses. The infima are finite because score zero gives

$$0 \leq \Psi(p_0, p_1) \leq \Phi(0)(p_0 + p_1), \qquad 0 \leq H(\bar{\Pi}) \leq \Phi(0) \sum_{i,\ell} \bar{\Pi}_{i\ell} = \tau B.$$

The final equality uses the row masses and $\sum_i q_i = 2$. Applying the nonnegative-variable rule to the class and coordinate prices gives

$$\begin{aligned} \inf_{\lambda\geq 0} \sum_k \lambda_k(\tau\theta_k - \mathbf{1}^\top\bar{\beta}_k) > -\infty \iff \mathbf{1}^\top\bar{\beta}_k \leq \tau\theta_k, \\ \inf_{w_0,w_1\geq 0} \sum_k w_k^\top[\bar{\beta}_k + \theta_k\bar{\mu} - A_k(\bar{\Pi})] > -\infty \iff A_k(\bar{\Pi}) \leq \bar{\beta}_k + \theta_k\bar{\mu}, \quad k = 0, 1. \end{aligned} \tag{B.7}$$

All linear terms then have infimum zero, leaving exactly $H(\bar{\Pi})$ as the dual objective. We can also eliminate $\bar{\beta}_k$: it must cover the positive movement left after subtracting $\theta_k\bar{\mu}$, while its total is at most $\tau\theta_k$. Coordinatewise,

$$\begin{aligned} \bar{\beta}_{kj} \geq 0, \quad A_{kj}(\bar{\Pi}) \leq \bar{\beta}_{kj} + \theta_k\bar{\mu}_j \iff \bar{\beta}_{kj} \geq (A_{kj}(\bar{\Pi}) - \theta_k\bar{\mu}_j)_+, \\ \mathbf{1}^\top\bar{\beta}_k \leq \tau\theta_k \implies \sum_j (A_{kj}(\bar{\Pi}) - \theta_k\bar{\mu}_j)_+ \leq \tau\theta_k. \end{aligned} \tag{B.8}$$

Conversely, when the last inequality holds, choosing $\bar{\beta}_k = (A_k(\bar{\Pi}) - \theta_k\bar{\mu})_+$ satisfies both price conditions. Therefore its elimination is exact, and the dual becomes

$$\begin{aligned} \max_{\tau,\bar{\Pi},\bar{\mu}\geq 0} \quad & H(\bar{\Pi}) \\ \text{subject to} \quad & \bar{\Pi}\mathbf{1} = \tau q, \qquad \tau + \bar{\mu}^\top z = 1, \\ & \sum_j (A_{kj}(\bar{\Pi}) - \theta_k\bar{\mu}_j)_+ \leq \tau\theta_k, \quad k = 0, 1. \end{aligned} \tag{B.9}$$

The remaining variables describe transported masses and their correction. Test scores and prices have all disappeared.

B.2 Show that the dual reaches the primal value

The derivation above gives a lower bound. To prove equality and dual attainment, we construct a strictly feasible point after removing coordinate prices that are necessarily zero. The primal minimum is attained by Appendix A.3; its existence argument used the bound B, but no constraint $\eta \leq B$ is added to (4).

A zero coordinate forces both its prices to vanish:

$$z_j = 0 \implies 0 \leq \theta_0 w_{0j} + \theta_1 w_{1j} \leq \eta z_j = 0 \implies w_{0j} = w_{1j} = 0.$$

Delete these fixed prices and the redundant inequalities they create. Choose $0 < \alpha < 1$ and use, on the remaining coordinates,

$$\begin{aligned} u = 0, \qquad a_i = \Phi(0) + 1, \qquad \lambda_k = 1, \qquad w_k = \alpha z, \\ J = 2[\Phi(0) + 1] + \theta_0 + \theta_1, \qquad \eta = J + 1 + \alpha(\theta_0 + \theta_1). \end{aligned} \tag{B.10}$$

These choices leave positive slack in every remaining constraint:

$$\begin{aligned} \Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i} &= -1 - \alpha\mathbf{d}_{i\ell}^\top z < 0, \\ w_{kj} - \lambda_k &= \alpha z_j - 1 < 0, \\ J - \eta &= -1 - \alpha(\theta_0 + \theta_1) < 0, \\ \theta_0 w_{0j} + \theta_1 w_{1j} - \eta z_j &= -(J+1)z_j < 0 \quad (z_j > 0). \end{aligned} \tag{B.11}$$

Moreover $w_{kj} > 0$ on retained coordinates and $\lambda_k > 0$. Thus finite-dimensional convex strong duality gives an attained dual maximum equal to $v(z)$ for the reduced problem.

Restore each deleted coordinate by setting

$$z_j = 0: \quad \bar{\beta}_{kj} = 0, \qquad \bar{\mu}_j = \max_k A_{kj}(\bar{\Pi})/\theta_k.$$

Then $\bar{\beta}_{kj} + \theta_k\bar{\mu}_j \geq A_{kj}(\bar{\Pi})$ for both classes, the $\bar{\beta}_k$ budgets do not increase, and $\bar{\mu}_j z_j = 0$ leaves normalization unchanged. Hence the full dual (B.9) also attains the value $v(z)$.

B.3 Normalize the masses to obtain the fraction

The rows of $\bar{\Pi}$ sum to τq. Dividing both $\bar{\Pi}$ and $\bar{\mu}$ by τ will restore the original source weights and remove z from their feasible set. The objective scales by the same factor: for $\tau > 0$,

$$\Psi(\tau p_0, \tau p_1) = \inf_s \tau[p_0\Phi(-s) + p_1\Phi(s)] = \tau\Psi(p_0, p_1),$$
$$H(\tau\Pi) = \sum_\ell \Psi(\tau p_{0\ell}, \tau p_{1\ell}) = \tau H(\Pi). \tag{B.12}$$

We must first ensure that an optimum has $\tau > 0$. If $\tau = 0$, its objective is zero:

$$\bar{\Pi}\mathbf{1} = 0 \implies \bar{\Pi} = 0 \implies H(\bar{\Pi}) = \sum_\ell \inf_s 0 = 0.$$

A positive feasible objective is obtained by sending every source to one target ℓ_*. With e_{ℓ_*} the vector having a single one at that target, let $\Pi = qe_{\ell_*}^\top$. The target then receives unit mass from each class. Convexity and score zero give

$$\Phi(-s) + \Phi(s) \geq 2\Phi(0), \qquad H(\Pi) = \Psi(1,1) = 2\Phi(0) = B > 0.$$

Correct all coordinate movements and scale back to the normalized dual by taking

$$\mu_j = \max_k \frac{A_{kj}(\Pi)}{\theta_k}, \qquad \tau = \frac{1}{1 + \mu^\top z}, \qquad \bar{\Pi} = \tau\Pi, \qquad \bar{\mu} = \tau\mu. \tag{B.13}$$
$$\bar{\Pi}\mathbf{1} = \tau q, \qquad \tau + \bar{\mu}^\top z = \tau(1 + \mu^\top z) = 1,$$
$$A_k(\Pi) - \theta_k\mu \leq 0 \implies (A_k(\bar{\Pi}) - \theta_k\bar{\mu})_+ = 0,$$
$$H(\bar{\Pi}) = \tau H(\Pi) = \tau B > 0.$$

This excludes $\tau = 0$ at an attained optimum. For any normalized feasible point with $\tau > 0$, define $\Pi = \bar{\Pi}/\tau$ and $\mu = \bar{\mu}/\tau$. Its constraints become

$$\bar{\Pi}\mathbf{1} = \tau q \iff \Pi\mathbf{1} = q,$$
$$A_k(\bar{\Pi}) = \tau A_k(\Pi),$$
$$(A_k(\bar{\Pi}) - \theta_k\bar{\mu})_+ = \tau(A_k(\Pi) - \theta_k\mu)_+, \tag{B.14}$$
$$\sum_j (A_{kj}(\bar{\Pi}) - \theta_k\bar{\mu}_j)_+ \leq \tau\theta_k \iff \sum_j (A_{kj}(\Pi) - \theta_k\mu_j)_+ \leq \theta_k.$$

For $\tau > 0$, positive-part scaling uses $(\tau t)_+ = \tau t_+$. These are precisely the certificate conditions (6), with no z. The one equation that still contains z determines the scale and hence the objective:

$$\tau + \bar{\mu}^\top z = 1 \iff \tau(1 + \mu^\top z) = 1 \iff \tau = (1 + \mu^\top z)^{-1},$$
$$H(\bar{\Pi}) = \tau H(\Pi) = \frac{H(\Pi)}{1 + \mu^\top z}. \tag{B.15}$$

Thus the denominator comes from restoring the row masses to q. It is positive because $\mu, z \geq 0$. Conversely, any pair satisfying (6) returns to the normalized dual by choosing this τ and setting $(\bar{\Pi}, \bar{\mu}) = \tau(\Pi, \mu)$. Equation (B.14) verifies its movement and row conditions, and $\tau(1 + \mu^\top z) = 1$ verifies normalization. The objective is the same in both directions, proving (7) and attainment of its maximum.

B.4 Connect the representation to selection and global cuts

For binary z, the risk model and the original finite problem have the same value. Starting from a feasible point of (2), set $\eta = J$ and $w_k = \lambda_k z$. Then

$$
\begin{aligned}
0 \le w_k &\le \lambda_k \mathbf{1}, \\
\theta_0 w_{0j} + \theta_1 w_{1j} &= z_j \theta^\top \lambda \le z_j J = z_j \eta, \\
a_i + \mathbf{d}_{i\ell}^\top w_{c_i} &= a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z.
\end{aligned}
\tag{B.16}
$$

The risk-price inequality (3) proves the middle line. Thus all risk-model constraints hold, and $v(z) \le V(z)$.

In the reverse direction, a feasible risk-model point satisfies

$$
\begin{aligned}
z_j = 0 &\implies \theta_0 w_{0j} + \theta_1 w_{1j} \le 0 \implies w_{0j} = w_{1j} = 0, \\
z_j = 1 &\implies w_{kj} \le \lambda_k, \\
&\implies \quad w_k \le \lambda_k z. \\
\Phi(y_i u_\ell) &\le a_i + \mathbf{d}_{i\ell}^\top w_{c_i} \le a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z.
\end{aligned}
\tag{B.17}
$$

Here $\theta_k > 0$ forces zero prices on unselected coordinates, and $\mathbf{d}_{i\ell} \ge 0$ proves the second inequality. Therefore (a, u, λ) is feasible in (2), with $V(z) \le J \le \eta$. Taking the minimum gives $V(z) \le v(z)$ and hence equality.

Finally, fix one feasible certificate pair (Π, μ). To obtain a global tangent, we need its fraction to be convex in z. Set $h = H(\Pi) \ge 0$, $r(z) = 1 + \mu^\top z > 0$, and $f(z) = h/r(z)$. Differentiating one coordinate at a time gives

$$
\begin{aligned}
\frac{\partial r}{\partial z_j} &= \mu_j, \qquad \frac{\partial f}{\partial z_j} = h[-r(z)^{-2}]\mu_j = -\frac{h\mu_j}{r(z)^2}, \\
\frac{\partial^2 f}{\partial z_m \partial z_j} &= -h\mu_j[-2r(z)^{-3}]\mu_m = \frac{2h\mu_j\mu_m}{r(z)^3}, \\
\nabla^2 f(z) &= \frac{2h\mu\mu^\top}{r(z)^3}.
\end{aligned}
\tag{B.18}
$$

For any direction $b \in \mathbb{R}^D$,

$$
b^\top \nabla^2 f(z) b = \frac{2h}{r(z)^3} (\mu^\top b)^2 \ge 0.
$$

Hence f is convex, and its tangent at any $z^s \in [0,1]^D$ satisfies

$$
f(z^s) + \nabla f(z^s)^\top (z - z^s) \le f(z) \le v(z).
$$

The certificate conditions do not depend on z, so (7) takes the maximum over a fixed family of convex functions; v is therefore convex as well. Together with binary equality and the attained fraction representation, this proves Theorem 2 and identifies the lower functions used by the algorithm.

C Proof of Theorem 3: valid bounds and finite termination

The proof follows the algorithm. First, each full solve supplies a cut and an actual test bound. Next, the filter forces any later evaluation to be sufficiently different from every earlier one. Counting these different selections proves finite termination.

C.1 A full solve supplies a reliable cut

At the evaluated subset z^s, use the feasible test bound R_s and certificate fraction f_s from Section 5. Write the allowed numerical downward shift as $e_s \geq 0$:

$$f_s(z) = \frac{h_s}{1 + (\mu^s)^\top z}, \qquad 0 \leq h_s \leq H(\Pi^s),$$
$$\ell_s = f_s(z^s) - e_s, \quad \ell_s \leq V(z^s) \leq R_s, \qquad R_s - \ell_s \leq \delta.$$

The shift is included in the inner gap δ; it does not change the fraction's gradient. Convexity and Theorem 2 give, for every $z \in [0,1]^D$,

$$\begin{aligned} g_s = \nabla f_s(z^s) &= -\frac{h_s \mu^s}{[1 + (\mu^s)^\top z^s]^2}, \\ L_s(z) &= \ell_s + g_s^\top (z - z^s) \\ &= f_s(z^s) + \nabla f_s(z^s)^\top (z - z^s) - e_s \\ &\leq f_s(z) - e_s \leq f_s(z) \leq v(z). \end{aligned}$$

Thus the cut is valid globally, even though its value and slope come from one fully evaluated subset. On binary subsets, $v = V$. Retaining the cuts and actual feasible tests therefore gives

$$a_t(z) = \max\{0, L_1(z), \ldots, L_t(z)\} \leq V(z), \qquad \mathrm{OPT} \leq U_t = \min\{B, R_1, \ldots, R_t\}.$$

C.2 The master certifies stopping or supplies a candidate

The master minimizes a_t over all feasible selections. Since $a_t \leq V$, its certified lower bound is also a lower bound on the true optimum:

$$\mathrm{LB}_t \leq \min_{z \in \mathcal{Z}_d} a_t(z) \leq \min_{z \in \mathcal{Z}_d} V(z) = \mathrm{OPT} \leq U_t.$$

Earlier master bounds remain valid when new cuts are added, because $a_{t+1} \geq a_t$. Taking the largest certified bound therefore preserves this interval. If $U_t - \mathrm{LB}_t \leq \varepsilon$, the retained test has risk at most $U_t \leq \mathrm{OPT} + \varepsilon$.

Otherwise, every accepted candidate must satisfy the algorithm's filter

$$a_t(z) \leq U_t - \varepsilon/2. \tag{C.1}$$

When the small pool has no such candidate, the master returns z_{M} with $a_t(z_{\mathrm{M}}) - \mathrm{LB}_t \leq \xi$. Using the updated lower bound and $\xi \leq \varepsilon/4$ gives

$$\begin{aligned} a_t(z_{\mathrm{M}}) &\leq \mathrm{LB}_t + \xi < U_t - \varepsilon + \xi \\ &\leq U_t - 3\varepsilon/4 < U_t - \varepsilon/2. \end{aligned}$$

Thus the master either certifies stopping or provides an eligible candidate. Prediction only orders eligible candidates and requires no accuracy assumption in this proof.

C.3 The filter forces evaluated subsets apart

The idea is that a cut is accurate at its own evaluated subset. A later candidate can pass the filter only if this cut drops by a definite amount. First, a previously evaluated z^s cannot be selected again:

$$a_t(z^s) \geq L_s(z^s) = \ell_s \geq R_s - \delta \geq U_t - \delta > U_t - \varepsilon/2.$$

To quantify how far the next subset must move, use the data weights M_j and distance ρ defined in Section 5. The row masses of a feasible certificate bound each movement coordinate:

$$\begin{aligned} 0 \leq A_{kj}(\Pi) &= \sum_{i \in I_k, \ell} \Pi_{i\ell} d_{i\ell,j} \\ &\leq \sum_{i \in I_k} \Big(\sum_{\ell} \Pi_{i\ell} \Big) \max_{\ell} d_{i\ell,j} \\ &= \sum_{i \in I_k} q_i \max_{\ell} d_{i\ell,j} \leq \theta_k M_j. \end{aligned}$$

If $\mu_j > M_j$, replacing it by M_j keeps $(A_{kj}(\Pi) - \theta_k \mu_j)_+ = 0$ for both classes. Clipping therefore preserves (6) and decreases the denominator. Hence we may use $0 \leq \mu^s \leq M$ before forming f_s and its cut. Choosing score zero in (5) also gives

$$0 \leq h_s \leq H(\Pi^s) \leq \Phi(0) \sum_{\ell} (p_{0\ell} + p_{1\ell}) = 2\Phi(0) = B,$$

$$0 \leq -g_{s,j} = \frac{h_s \mu_j^s}{[1 + (\mu^s)^\top z^s]^2} \leq B M_j.$$

Now consider an earlier evaluation $s < t$ and the subset z^t accepted after $t-1$ completed evaluations. The retained test gives $U_{t-1} \leq R_s$, and the retained cut gives $L_s \leq a_{t-1}$. The inner accuracy and filter thus imply

$$\ell_s \geq R_s - \delta \geq U_{t-1} - \delta, \qquad L_s(z^t) \leq a_{t-1}(z^t) \leq U_{t-1} - \varepsilon/2.$$

Subtract and use the bounded nonpositive gradient:

$$\begin{aligned} \varepsilon/2 - \delta \leq \ell_s - L_s(z^t) &= -g_s^\top (z^t - z^s) \\ &\leq \sum_j (-g_{s,j})(z_j^t - z_j^s)_+ \\ &\leq B \sum_j M_j (z_j^t - z_j^s)_+ \leq B \rho(z^t, z^s). \end{aligned}$$

In the second line, removing negative contributions can only increase the sum. Consequently every pair of completed evaluations satisfies

$$\rho(z^t, z^s) \geq r_\varepsilon = \frac{\varepsilon/2 - \delta}{B} > 0.$$

C.4 Count the distinct patterns and conclude

Only the largest coordinate weights are needed to distinguish such separated subsets. Sort $M_{(1)} \geq \cdots \geq M_{(D)}$ and retain the first s_ε coordinates, where

$$s_\varepsilon = \min \left\{ s \in \{0, \ldots, D\} : \sum_{j > s} M_{(j)} < r_\varepsilon \right\}.$$

If two evaluated subsets had the same binary pattern on these coordinates, they could differ only on the omitted coordinates. Write $z_{(j)}$ for the entry at the coordinate with weight $M_{(j)}$. Since binary entries differ by at most one,

$$\rho(z, z') = \sum_{j>s_\varepsilon} M_{(j)} |z_{(j)} - z'_{(j)}| \le \sum_{j>s_\varepsilon} M_{(j)} < r_\varepsilon,$$

contradicting separation. The tail inequality is strict because the separation bound allows equality.

Each completed evaluation therefore has a different retained pattern. Selecting exactly i retained coordinates gives $\binom{s_\varepsilon}{i}$ patterns, and the budget allows at most d. Hence

$$T_\varepsilon \le \min\left\{ |\mathcal{Z}_d|, \sum_{i=0}^{\min(d, s_\varepsilon)} \binom{s_\varepsilon}{i} \right\} \le 2^{s_\varepsilon}.$$

If the algorithm did not stop, the pool or master would supply another eligible candidate. Completing its solve would require a further distinct pattern, so continuation cannot exceed this finite count. It follows that $U_t - \mathrm{LB}_t \le \varepsilon$ is reached and the returned test has risk at most OPT + ε.

T_ε includes the first full evaluation; numerical refinements at one selected subset count as part of that evaluation. The conclusion assumes the required inner and master accuracies are attained. A premature budget exit instead returns the available feasible test and valid bounds. All numerical certificate and comparison errors are included in the stated accuracy budgets.

D Loss formulas for Section 6

Every row in Section 6 is obtained by minimizing the same scalar objective. Write $F(s) = p_0\Phi(-s) + p_1\Phi(s)$, so $\Psi(p_0, p_1) = \inf_s F(s)$. First take $p_0, p_1 > 0$; the axes are treated below. When the minimum is attained, s^* denotes a minimizing scalar score.

For half hinge, $\Phi(s) = (1+s)_+/2$. If $s > 1$, replacing it by 1 keeps $\Phi(-s) = 0$ and reduces $\Phi(s)$. If $s < -1$, replacing it by -1 keeps $\Phi(s) = 0$ and reduces $\Phi(-s)$. It is therefore sufficient to minimize on $[-1, 1]$:

$$\begin{aligned} F(s) &= \frac{p_0(1-s) + p_1(1+s)}{2} = \frac{p_0 + p_1}{2} + \frac{p_1 - p_0}{2} s, \\ \Psi(p_0, p_1) &= \begin{cases} F(-1) = p_0, & p_1 \ge p_0, \\ F(1) = p_1, & p_1 < p_0, \end{cases} = \min(p_0, p_1). \end{aligned} \tag{D.1}$$

The choice of endpoint follows from the sign of the linear coefficient; when $p_0 = p_1$, every score in the interval is optimal.

For logistic, $\Phi(s) = \log(1 + e^s)$. Differentiate the two terms separately:

$$\begin{aligned} \frac{d}{ds}\left[p_0 \log(1 + e^{-s})\right] &= -\frac{p_0 e^{-s}}{1 + e^{-s}} = -\frac{p_0}{1 + e^s}, \\ \frac{d}{ds}\left[p_1 \log(1 + e^{s})\right] &= \frac{p_1 e^s}{1 + e^s}, \\ F'(s) = \frac{-p_0 + p_1 e^s}{1 + e^s} = 0 \quad &\Leftrightarrow \quad -p_0 + p_1 e^{s^*} = 0 \\ &\Leftrightarrow \quad e^{s^*} = p_0/p_1, \quad s^* = \log(p_0/p_1). \end{aligned}$$

The derivative changes from negative to positive at s^*, so this is the minimum. Substituting both logarithms gives

$$\begin{aligned} 1 + e^{-s^*} &= (p_0 + p_1)/p_0, \qquad & 1 + e^{s^*} &= (p_0 + p_1)/p_1, \\ \Psi(p_0, p_1) &= p_0 \log \frac{p_0 + p_1}{p_0} + p_1 \log \frac{p_0 + p_1}{p_1} \\ &= p_0[\log(p_0 + p_1) - \log p_0] + p_1[\log(p_0 + p_1) - \log p_1] \\ &= (p_0 + p_1)\log(p_0 + p_1) - p_0 \log p_0 - p_1 \log p_1. \end{aligned} \tag{D.2}$$

For exponential, $\Phi(s) = e^s$. A square proves the lower bound and identifies equality:

$$\begin{aligned} F(s) - 2\sqrt{p_0 p_1} &= \left(\sqrt{p_0} e^{-s/2} - \sqrt{p_1} e^{s/2}\right)^2 \geq 0, \\ \sqrt{p_0} e^{-s^*/2} = \sqrt{p_1} e^{s^*/2} \quad &\Leftrightarrow \quad e^{s^*} = \sqrt{p_0/p_1}, \quad s^* = \tfrac{1}{2}\log(p_0/p_1), \\ \Psi(p_0, p_1) &= F(s^*) = 2\sqrt{p_0 p_1}. \end{aligned} \tag{D.3}$$

For squared hinge, $\Phi(s) = (1 + s)_+^2$. The same clipping argument restricts the score to $[-1, 1]$. Expanding and completing the square gives

$$\begin{aligned} F(s) &= p_0(1 - s)^2 + p_1(1 + s)^2 \\ &= (p_0 + p_1)s^2 + 2(p_1 - p_0)s + (p_0 + p_1) \\ &= (p_0 + p_1)\left(s - \frac{p_0 - p_1}{p_0 + p_1}\right)^2 + (p_0 + p_1) - \frac{(p_0 - p_1)^2}{p_0 + p_1}, \\ (p_0 + p_1) - \frac{(p_0 - p_1)^2}{p_0 + p_1} &= \frac{(p_0 + p_1)^2 - (p_0 - p_1)^2}{p_0 + p_1} = \frac{4p_0 p_1}{p_0 + p_1}, \\ s^* &= \frac{p_0 - p_1}{p_0 + p_1} \in [-1, 1], \qquad \Psi(p_0, p_1) = F(s^*) = \frac{4p_0 p_1}{p_0 + p_1}. \end{aligned} \tag{D.4}$$

The nonnegative square vanishes at this feasible score, which proves the minimum.

All four losses in this appendix satisfy $\Phi(s) \to 0$ as $s \to -\infty$. Thus zero mass from one class lets its opposite score direction drive the remaining loss to zero. Nonnegativity supplies the matching lower bound:

$$\begin{aligned} 0 \leq \Psi(p_0, 0) \leq p_0 \Phi(-s) \longrightarrow 0 \quad (s \to +\infty), \\ 0 \leq \Psi(0, p_1) \leq p_1 \Phi(s) \longrightarrow 0 \quad (s \to -\infty), \\ \Psi(p_0, 0) = \Psi(0, p_1) = \Psi(0, 0) = 0. \end{aligned}$$

For half hinge and squared hinge, the first zero is already attained at $s \geq 1$, and the second at $s \leq -1$. For logistic and exponential with a positive remaining mass, it is only an infimum as the score tends in the stated direction. The closed formulas use $0 \log 0 = 0$ and assign the squared-hinge value zero when both masses are zero.

E Risk predictions for the selection rule

A completed inner evaluation has the certified gap $R_s - \ell_s \leq \delta$, so

$$0 \leq R_s - V(z^s) \leq R_s - \ell_s \leq \delta.$$

Thus each completed solve provides an approximate observation of the best risk for its binary subset. We use these observations to rank the subsets passing the global-cut filter.

E.1 Predicting risk and uncertainty

Normalize risks by B and define the similarity kernel

$$r_t = (R_1/B, \dots, R_t/B)^\top,$$
$$k(z, z') = \tfrac{1}{4} \exp\left(-\frac{\|z - z'\|_1}{\max(1, d)}\right), \qquad \nu = \max\{10^{-8}, (\delta/B)^2\}.$$

Here r_t collects the normalized observed risks. The kernel k decreases with the number of coordinates on which two subsets differ. The fixed parameter ν stabilizes the computation; it does not assume a probability law for the inner-solve errors. Let I_t be the identity matrix and $\mathbf{1}_t$ the vector of ones. Define

$$K_t = (k(z^s, z^r))_{s,r=1}^t, \qquad k_t(z) = (k(z^1, z), \dots, k(z^t, z))^\top, \qquad A_t = K_t + \nu I_t.$$

The matrix K_t stores similarities among evaluated subsets, and the vector $k_t(z)$ compares a proposal with those subsets. To obtain a prediction, let Y model the normalized observations and F the proposal's normalized risk. Our calculation seeks

$$m_t(z) = B\,\mathbb{E}[F \mid Y = r_t], \qquad \sigma_t(z)^2 = B^2 \operatorname{Var}(F \mid Y = r_t).$$

Use prior mean $1/2$ and the following joint Gaussian model. Positivity of its covariance matrix is verified below.

$$\begin{pmatrix} Y \\ F \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \frac{1}{2}\mathbf{1}_t \\ \frac{1}{2} \end{pmatrix}, \begin{pmatrix} A_t & k_t(z) \\ k_t(z)^\top & \frac{1}{4} \end{pmatrix}\right).$$

Subtract a linear prediction based on Y from F. Choosing its coefficient as $k_t(z)^\top A_t^{-1}$ cancels the covariance with Y, leaving the residual

$$E = F - \tfrac{1}{2} - k_t(z)^\top A_t^{-1}(Y - \tfrac{1}{2}\mathbf{1}_t).$$

The residual has mean zero. Its covariance with Y and its own variance are

$$\begin{aligned} \operatorname{Cov}(E, Y) &= k_t(z)^\top - k_t(z)^\top A_t^{-1} A_t = 0, \\ \operatorname{Var}(E) &= \tfrac{1}{4} - 2k_t(z)^\top A_t^{-1} k_t(z) + k_t(z)^\top A_t^{-1} A_t A_t^{-1} k_t(z) \\ &= \tfrac{1}{4} - k_t(z)^\top A_t^{-1} k_t(z). \end{aligned}$$

Because (E, Y) is jointly Gaussian, zero covariance gives independence. Conditioning on the completed observations therefore leaves the mean and variance of E unchanged:

$$\begin{aligned} \mathbb{E}[F \mid Y = r_t] &= \tfrac{1}{2} + k_t(z)^\top A_t^{-1}(r_t - \tfrac{1}{2}\mathbf{1}_t), \\ \operatorname{Var}(F \mid Y = r_t) &= \operatorname{Var}(E) = \tfrac{1}{4} - k_t(z)^\top A_t^{-1} k_t(z). \end{aligned}$$

Returning to the original risk scale gives

$$\begin{aligned} m_t(z) &= B\left[\tfrac{1}{2} + k_t(z)^\top A_t^{-1}(r_t - \tfrac{1}{2}\mathbf{1}_t)\right], \\ \sigma_t(z) &= B\sqrt{\tfrac{1}{4} - k_t(z)^\top A_t^{-1} k_t(z)}. \end{aligned} \tag{E.1}$$

For $t = 0$, use $m_0 = B/2$ and $\sigma_0 = B/2$. These are model predictions for the selection rule, not certified confidence bounds. The global guarantee in Theorem 3 does not assume that the model describes the true risk correctly.

To verify the covariance used above, each binary coordinate contributes the matrix with positive eigenvalues:

$$\begin{pmatrix} 1 & e^{-1/\max(1,d)} \\ e^{-1/\max(1,d)} & 1 \end{pmatrix}, \qquad 1 \pm e^{-1/\max(1,d)} > 0.$$

The kernel matrix on all binary subsets is the tensor product of these matrices, scaled by $1/4$. Hence the kernel matrix for any finite list of subsets is positive semidefinite. In the joint covariance above, νI_t adds a positive quadratic-form contribution whenever the observation component is nonzero; a vector with only its last component nonzero has positive variance $1/4$. The joint matrix is therefore positive definite. This makes A_t invertible and its conditional variance strictly positive.

E.2 Updating the predictor with little overhead

Maintain the lower-triangular Cholesky factor C_t of A_t. One new completed observation adds one row and column:

$$A_t = C_t C_t^\top, \qquad A_{t+1} = \begin{pmatrix} A_t & k_t(z^{t+1}) \\ k_t(z^{t+1})^\top & \frac{1}{4} + \nu \end{pmatrix}.$$

Retain C_t and seek a new lower-triangular factor with row $v_*^\top$ and positive diagonal c. Block multiplication gives

$$\begin{aligned} C_{t+1}C_{t+1}^\top &= \begin{pmatrix} C_t & 0 \\ v_*^\top & c \end{pmatrix}\begin{pmatrix} C_t^\top & v_* \\ 0 & c \end{pmatrix} \\ &= \begin{pmatrix} A_t & C_t v_* \\ v_*^\top C_t^\top & v_*^\top v_* + c^2 \end{pmatrix}. \end{aligned}$$

Matching this matrix to A_{t+1} determines the new entries:

$$\begin{aligned} C_t v_* &= k_t(z^{t+1}), \\ v_*^\top v_* + c^2 = \tfrac{1}{4} + \nu \quad &\Longrightarrow \quad c = \sqrt{\tfrac{1}{4} + \nu - v_*^\top v_*}. \end{aligned}$$

The square-root argument is positive because A_{t+1} is positive definite. Thus the update needs one triangular solve and one appended row. For predictions, compute the observation weights α_t using two triangular solves, and compute v for each proposal:

$$C_t p = r_t - \tfrac{1}{2}\mathbf{1}_t, \qquad C_t^\top \alpha_t = p, \qquad C_t v = k_t(z).$$

Here p is a forward-solve intermediate. The first two equations give $A_t\alpha_t = r_t - \mathbf{1}_t/2$; the last gives $k_t(z)^\top A_t^{-1} k_t(z) = v^\top v$. Substitution into (E.1) yields

$$m_t(z) = B[\tfrac{1}{2} + k_t(z)^\top \alpha_t], \qquad \sigma_t(z) = B\sqrt{\max\{0, \tfrac{1}{4} - v^\top v\}}.$$

Updating the factor and predicting a bounded batch cost $O(t^2)$ arithmetic operations, with $O(Dt)$ additional work for each similarity vector. The kernel parameters remain fixed, so no hyperparameter optimization is needed between inner solves. The maximum with zero only removes numerical roundoff.